

Day 9 Plan

Maths: Time & Work, Pipes & Cisterns · Reasoning: Order & Ranking, Dictionary Order · Science: Chemistry 1: Matter, Mixtures, Atoms & Molecules

Today's schedule (about 5½ hours with breaks)

#	Block	Time	What you do	Where
0	Day 8 recall	15 min	10 questions from Day 8, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	5 simplification questions for speed → formulas & methods (10 min) → 2-3 book questions per pattern, timed → 10 Testbook PYQs on this topic → log mistakes	Section A · book · Testbook
-	Break	10 min	Away from the screen	-
2	Reasoning	45 min	Learn the method (10 min) → 15 timed questions, Testbook PYQs first → log	Section B · Testbook
-	Break	10 min	Away from the screen	-
3	Science	110 min	8 lessons with quick checks (≈60 min) → 20-question practice set → Testbook PYQs: matter, atoms, molecules, formulae (S tier — chemistry is the biggest science share)	Section C · Testbook
-	Break	10 min	Away from the screen	-
4	General awareness	45 min	20 current-affairs MCQs (20 min) → static GA: Polity 3 — President, Parliament, judiciary, key amendments — Testbook GA PYQs on it (25 min)	CA PDF · Testbook
5	Close	15 min	Redo today's wrong questions · write 3 rules for tomorrow · send me your scores	Mistake log

Short day? Cut in this order: Testbook science PYQs → reasoning down to 10 questions → static GA down to 10 min. Never cut recall, maths or the science lessons. · Exam pace (average 54 sec a question): GA ~30 sec · Science ~45 · Reasoning ~50 · Maths ~65.

Block 0 — Day 8 recall (answers on the last page)

#	Question	Your answer
1	SI unit of force?	
2	Value of g on Earth's surface?	
3	Mirror used by dentists?	
4	Escape velocity from Earth?	
5	1 horsepower = ? watt	
6	Lens used to correct hypermetropia?	
7	6 Ω and 3 Ω in parallel = ?	
8	Which is a vector: speed or velocity?	
9	Unit of electrical energy used in bills?	
10	Maths: +10% then +10% = net ?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Time & Work, Pipes & Cisterns

Core idea

Treat work as a **total number of units**, not as a fraction. Take the total work = **LCM of the days** given; then each person's **efficiency** (units per day) = total ÷ their days. Everything becomes whole-number addition. Pipes are the same idea: a filling pipe adds units, a leak or outlet pipe subtracts them.

Formulas

Concept	Formula / rule
One day's work	A finishes in a days → does $1/a$ of the work per day
A and B together	$ab/(a + b)$ days
LCM method	Total = LCM of days · efficiency = total ÷ days · time = total ÷ combined efficiency
Efficiency vs time	Inversely proportional: twice as efficient → half the time
Men-days-hours	$M_1D_1H_1/W_1 = M_2D_2H_2/W_2$
Wages	Shared in the ratio of work done (efficiency × days worked)
Pipes	Filling pipe = +, emptying pipe / leak = –
A + B, B + C, C + A given	A + B + C = half the sum of the three pair-rates

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Two together	LCM	A 10 days, B 15 days → total 30; A 3/day, B 2/day → $30 ÷ 5 = \mathbf{6 \text{ days}}$
2. One leaves	Work done, then the rest	A 12, B 18 → total 36 (A 3, B 2). Together 4 days = 20 → B does 16 more ÷ 2 = 8 days
3. Efficiency given	Use ratio as units	A is twice as efficient as B; together 14 days → total = $3 \times 14 = 42$ → B alone 42 days
4. Pairs given	Add all pairs, halve	A+B 12, B+C 15, C+A 20 → total 60 → 5, 4, 3 → A+B+C = 6 → 10 days ; A = 6 – 4 = 2 → A alone 30 days
5. Men and days	$M_1D_1 = M_2D_2$	12 men take 18 days → 27 men take $12 \times 18 \div 27 = \mathbf{8 \text{ days}}$
6. Wages	Ratio of efficiency	A 10 days, B 15 days, paid ₹1500 together → 3 : 2 → A ₹900, B ₹600
7. Fill + leak	+ and –	A fills in 10 h, B empties in 15 h → total 30: $+3 - 2 = 1/h$ → 30 h
8. Two fill, one empties	Net rate	A 12 h, B 15 h fill; C empties in 20 h → total 60: $5 + 4 - 3 = 6$ → 10 h
9. Find the leak	Difference of rates	Fills in 6 h, with a leak in 8 h → total 24: $4 - 3 = 1$ → leak alone empties in 24 h

Traps

- Wages follow **work done**, not the number of days alone.
- A leak's rate is **negative**. Don't add it.
- 'Half the work' or '2/3 of the job' — scale the total units first.

B · REASONING — Order & Ranking, Dictionary Order

Method: draw a line of positions and mark people on it. One formula answers most questions:
Total = rank from top + rank from bottom – 1.

Find	Formula
Total persons	Left rank + right rank – 1
Rank from the other end	Total – rank + 1
Persons between A and B (no overlap)	Total – (A's rank + B's rank), counted from opposite ends
Swap questions	After swapping, the new rank of one person + the other's old rank from the other end – 1 = total

Dictionary order: compare words letter by letter; the first different letter decides. A shorter word that is the start of a longer one comes first (PEN before PENCIL).

Patterns, with solved examples

Pattern	Example	Solution
Total	Ram is 12th from the top and 18th from the bottom. How many students?	$12 + 18 - 1 = \mathbf{29}$
Other end	In a row of 40, A is 15th from the left. Position from the right?	$40 - 15 + 1 = \mathbf{26th}$
Between	Class of 50: A is 10th from the top, B is 20th from the bottom. How many between them?	$50 - (10 + 20) = \mathbf{20}$
Swap	A is 8th from the left, B is 12th from the right. They swap; A is now 15th from the left. Total?	$15 + 12 - 1 = \mathbf{26}$
Dictionary	Arrange: Pencil, Pen, Pendant, Penalty	Pen, Penalty, Pencil, Pendant (4th letters a, c, d)
Position of a word	Which comes 3rd: Train, Trail, Track, Trade?	Track, Trade, Trail, Train → Trail

Speed rule

Draw the row. Most ranking mistakes are off-by-one errors — the '– 1' and '+ 1' in the formulas.

C · SCIENCE — Chemistry 1: Matter, Mixtures, Atoms & Molecules

8 lessons — the foundation for all chemistry. Chemistry was the largest science share in the 2026 papers (R: 81 of 210), and atoms, molecules and formulae alone were 22 of them. Learn the symbol/valency table by heart.

Lesson 1 · States of matter and changes of state (about 7 min)

Matter is anything that has mass and occupies space. Its particles are always moving and attract each other. The three common states differ in how tightly the particles are held.

Property	Solid	Liquid	Gas
Shape / volume	Fixed / fixed	Not fixed / fixed	Not fixed / not fixed
Compressibility	Negligible	Very low	High (LPG, CNG cylinders)
Particle spacing, force	Closest, strongest	Middle	Far apart, weakest
Diffusion	Slowest	Faster	Fastest

Changes of state: **melting** (solid → liquid), **boiling/evaporation** (liquid → gas), **condensation** (gas → liquid), **freezing**, and **sublimation** (solid → gas directly). Substances that sublime: **camphor, naphthalene, iodine, ammonium chloride, dry ice (solid CO₂)**.

Evaporation happens at any temperature, only at the surface, and causes **cooling** (an earthen pot cools water; sweating cools us). It increases with temperature, surface area and wind, and decreases with humidity. Temperature in kelvin: **K = °C + 273**. The 4th and 5th states: **plasma** (stars, neon signs) and **Bose-Einstein condensate**.

Quick check 1 — answer out loud before moving on (answers at the end)

a) Name two substances that sublime. b) Why does water in an earthen pot stay cool? c) 25 °C = ? K

Lesson 2 · Pure substances and mixtures; solutions, colloids, suspensions (about 9 min)

Type	Key idea	Examples
Element	One kind of atom	Iron, oxygen, carbon
Compound	Elements in a fixed ratio ; new properties; separated only chemically	Water (H : O = 1 : 8 by mass), salt, CO ₂
Mixture	Any ratio; components keep their properties; separated physically	Air, brass, sea water, soil

Mixtures can be **homogeneous** (uniform — salt water, air, alloys) or **heterogeneous** (sand in water, oil in water).

	True solution	Colloid	Suspension
Particle size	< 1 nm	≈ 1–100 nm (some books 1–1000)	> 100 nm
Settles?	No	No	Yes
Tyndall effect (scatters a beam)	No	Yes	Yes

	True solution	Colloid	Suspension
Example	Salt or sugar in water	Milk, fog, smoke, jelly	Muddy water, chalk in water
Colloid type	Dispersed in	Examples	
Aerosol	Liquid or solid in gas	Fog, clouds, mist · smoke, exhaust	
Emulsion	Liquid in liquid	Milk, face cream	
Foam	Gas in liquid / solid	Shaving cream · foam rubber, sponge, pumice	
Sol / solid sol	Solid in liquid / solid	Milk of magnesia, mud · coloured gemstone, milky glass	
Gel	Liquid in solid	Jelly, cheese, butter	

Exam trap

Milk is a **colloid** (emulsion), not a solution. Sunlight streaming through a dense forest or a dusty room is the **Tyndall effect**.

Quick check 2 — answer out loud before moving on (answers at the end)

a) Is brass an element, compound or mixture? b) Which shows the Tyndall effect: salt solution or milk?

Lesson 3 · Separating mixtures (about 7 min)

Method	Use it when	Example
Evaporation	A dissolved solid from a liquid	Salt from sea water
Crystallisation	Pure crystals (removes soluble impurities — better than evaporation)	Pure copper sulphate, alum
Centrifugation	Very fine suspended particles, spun fast	Cream from milk; blood tests; washing-machine dryer
Separating funnel	Two liquids that don't mix	Oil and water
Sublimation	One component sublimates	Ammonium chloride or camphor from salt
Chromatography	Coloured components in small amounts	Dyes in ink; pigments; drugs in blood
Distillation	Miscible liquids, boiling points differ by > 25 K	Acetone and water
Fractional distillation	Boiling points differ by < 25 K	Gases from liquid air; petroleum fractions
Magnet · sieving · winnowing	Magnetic part · size · light husk	Iron filings · flour · grain from husk
Loading / sedimentation	Make particles settle	Alum added to muddy water

Quick check 3 — answer out loud before moving on (answers at the end)

a) How is cream separated from milk? b) How are oxygen and nitrogen obtained from air?

Lesson 4 · Physical vs chemical changes (about 4 min)

	Physical change	Chemical change
New substance?	No	Yes
Usually reversible?	Yes	No
Examples	Melting ice, dissolving sugar or salt, breaking glass, magnetising iron, boiling water	Rusting, burning, curdling of milk, ripening of fruit, cooking, digestion, photosynthesis, respiration

Exam trap

A burning candle shows **both**: the wax melting is physical, the wax burning is chemical.

Quick check 4 — answer out loud before moving on (answers at the end)

Classify: a) rusting of iron b) melting of wax c) curdling of milk.

Lesson 5 · Atoms, laws of chemical combination and the mole

(about 8 min)

Dalton's atomic theory: matter is made of tiny indivisible atoms; atoms of one element are identical; atoms combine in small whole-number ratios. (We now know atoms are divisible — protons, neutrons, electrons.)

Law	Given by	Statement
Conservation of mass	Lavoisier	Mass is neither created nor destroyed in a chemical reaction
Constant (definite) proportions	Proust	A compound always contains the same elements in the same mass ratio (water: H : O = 1 : 8)

Mole: 1 mole = 6.022×10^{23} particles (Avogadro's number) = the molar mass in grams = **22.4 L** for a gas at STP. Moles = given mass $\div$ molar mass.

Worked example — 36 g of water

Molar mass of H_2O = $2 \times 1 + 16 = 18$ g

Moles = $36 \div 18 = 2$ mol

Molecules = $2 \times 6.022 \times 10^{23} = 1.2044 \times 10^{24}$

Quick check 5 — answer out loud before moving on (answers at the end)

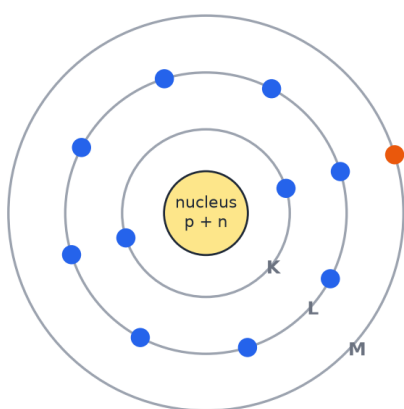
a) Mass of 1 mole of CO_2 ? b) Who gave the law of conservation of mass?

Lesson 6 · Inside the atom (about 9 min)

Particle	Charge	Discovered by	Note
Electron	-1	J. J. Thomson (cathode rays)	Mass $\approx 1/1840$ of a hydrogen atom
Proton	+1	E. Goldstein (canal rays); named by Rutherford	In the nucleus
Neutron	0	James Chadwick (1932)	In the nucleus (absent in ordinary hydrogen)

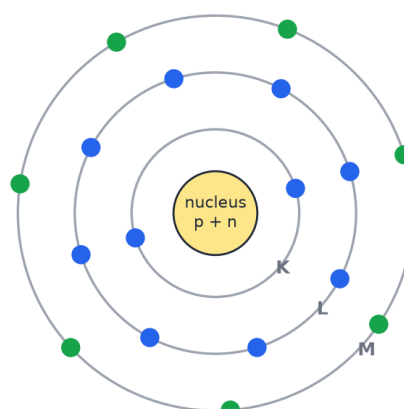
Rutherford's gold-foil experiment (α -particles): most went straight through $\rightarrow$ the atom is **mostly empty space**; a few bounced back $\rightarrow$ a tiny, dense, **positive nucleus**. **Bohr**: electrons move in fixed shells K, L, M, N; a shell holds at most **$2n^2$** electrons (2, 8, 18, 32), and the outermost shell holds at most 8.

Sodium (Na) · Z = 11
configuration 2, 8, 1



1 outer electron $\rightarrow$ loses it $\rightarrow$ Na^+ (valency 1)

Chlorine (Cl) · Z = 17
configuration 2, 8, 7



7 outer electrons $\rightarrow$ gains 1 $\rightarrow$ Cl^- (valency 1)

Atomic number Z = number of protons. **Mass number A** = protons + neutrons. **Isotopes**: same Z, different A (^1H protium, ^2H deuterium, ^3H tritium; C-12 and C-14). **Isobars**: same A, different Z (^{40}Ar and ^{40}Ca).

Isotope	Use
Uranium-235	Nuclear reactor fuel
Cobalt-60	Cancer treatment (radiotherapy)
Iodine-131	Goitre / thyroid treatment
Carbon-14	Dating fossils and old objects

Quick check 6 — answer out loud before moving on (answers at the end)

a) Maximum electrons in the M shell? b) Which isotope is used to treat goitre? c) Who discovered the neutron?

Lesson 7 • Electronic configuration and valency (about 7 min)

Valency = electrons an atom loses, gains or shares to complete its outer shell (8, or 2 for the first shell). Outer electrons 1–3 → valency = that number; 4 → 4; 5–7 → 8 minus that number; full shell → 0.

Element	Z	Configuration	Valency	Element	Z	Configuration	Valency
H	1	1	1	Na	11	2, 8, 1	1
He	2	2	0	Mg	12	2, 8, 2	2
C	6	2, 4	4	Al	13	2, 8, 3	3
N	7	2, 5	3	Si	14	2, 8, 4	4
O	8	2, 6	2	P	15	2, 8, 5	3 (or 5)
F	9	2, 7	1	S	16	2, 8, 6	2
Ne	10	2, 8	0	Cl	17	2, 8, 7	1
K	19	2, 8, 8, 1	1	Ca	20	2, 8, 8, 2	2

Quick check 7 — answer out loud before moving on (answers at the end)

a) Configuration of Mg? b) Valency of nitrogen?

Lesson 8 • Ions and writing chemical formulae (about 9 min)

Valency	Positive ions	Negative ions
1	Na ⁺ , K ⁺ , Ag ⁺ , NH ₄ ⁺ (ammonium), H ⁺	Cl ⁻ , OH ⁻ (hydroxide), NO ₃ ⁻ (nitrate), HCO ₃ ⁻ (hydrogen carbonate)
2	Mg ²⁺ , Ca ²⁺ , Zn ²⁺ , Cu ²⁺ , Fe ²⁺	O ²⁻ , S ²⁻ , SO ₄ ²⁻ (sulphate), CO ₃ ²⁻ (carbonate)
3	Al ³⁺ , Fe ³⁺	PO ₄ ³⁻ (phosphate), N ³⁻

Criss-cross method: write the symbols, swap the valencies down as subscripts, cancel any common factor, and put a group of atoms in brackets.

Worked example — writing formulae

Aluminium oxide: Al(3) O(2) → **Al₂O₃**

Calcium hydroxide: Ca(2) OH(1) → **Ca(OH)₂**

Ammonium sulphate: NH₄(1) SO₄(2) → **(NH₄)₂SO₄**

Magnesium oxide: Mg(2) O(2) → Mg₂O₂ → cancel → **MgO**

Molecule	Molecular mass	Molecule	Molecular mass
H ₂ O	18	NH ₃	17
CO ₂	44	NaCl	58.5
CaCO ₃	100	H ₂ SO ₄	98
HCl	36.5	Glucose C ₆ H ₁₂ O ₆	180

Atomicity = atoms in one molecule: He 1 (monatomic), O₂ 2, O₃ (ozone) 3, P₄ 4, S₈ 8. Atomic masses to know: H 1, C 12, N 14, O 16, Na 23, Mg 24, Al 27, S 32, Cl 35.5, K 39, Ca 40.

Quick check 8 — answer out loud before moving on (answers at the end)

a) Formula of aluminium sulphate? b) Molecular mass of H₂SO₄?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
States	Sublime: camphor, naphthalene, iodine, NH_4Cl , dry ice · evaporation cools · $\text{K} = ^\circ\text{C} + 273$
Mixtures	Compound = fixed ratio · colloid 1-100 nm, Tyndall · milk = emulsion, fog = aerosol, jelly = gel
Separation	Cream → centrifuge · oil/water → funnel · dyes → chromatography · air gases, petroleum → fractional distillation
Changes	Chemical: rusting, burning, curdling, ripening, digestion · physical: melting, dissolving
Laws	Lavoisier: conservation of mass · Proust: constant proportion (water $\text{H} : \text{O} = 1 : 8$)
Mole	6.022×10^{23} · 22.4 L at STP · moles = mass ÷ molar mass
Atom	e^- Thomson · p^+ Goldstein · n Chadwick 1932 · nucleus Rutherford · shells $2n^2$
Isotopes	Same Z · U-235 fuel · Co-60 cancer · I-131 goitre · C-14 dating · isobars: same A
Formulae	Al_2O_3 , $\text{Ca}(\text{OH})_2$, $(\text{NH}_4)_2\text{SO}_4$, Na_2CO_3 · H_2SO_4 98, CaCO_3 100, CO_2 44, NaCl 58.5

Science practice set — 20 questions · target 6 minutes

#	Question
1	Which of these sublimes on heating? (a) Common salt (b) Camphor (c) Sugar (d) Wax
2	Evaporation causes: (a) cooling (b) heating (c) no temperature change (d) condensation
3	100 °C in kelvin is: (a) 273 K (b) 100 K (c) 173 K (d) 373 K
4	Brass is a(n): (a) mixture (alloy) (b) element (c) compound (d) colloid
5	Milk is an example of a(n): (a) true solution (b) suspension (c) aerosol (d) emulsion
6	The Tyndall effect is shown by: (a) true solutions only (b) colloids (c) pure water (d) pure elements
7	Fog is a colloid of: (a) gas in liquid (b) liquid in gas (c) solid in gas (d) liquid in solid
8	Cream is separated from milk by: (a) filtration (b) centrifugation (c) sublimation (d) chromatography
9	Different gases are obtained from air by: (a) evaporation (b) sublimation (c) fractional distillation (d) crystallisation
10	Dyes in black ink are separated by: (a) chromatography (b) distillation (c) filtration (d) decantation
11	Which is a chemical change? (a) Rusting of iron (b) Melting of ice (c) Dissolving sugar (d) Breaking glass
12	The law of constant proportions was given by: (a) Lavoisier (b) Proust (c) Dalton (d) Avogadro
13	Number of moles in 88 g of CO ₂ : (a) 1 (b) 2 (c) 4 (d) 0.5
14	The neutron was discovered by: (a) Chadwick (b) Thomson (c) Rutherford (d) Goldstein
15	Rutherford's experiment showed that the atom has a: (a) uniform positive sphere (b) negative nucleus (c) no empty space (d) small, dense, positive nucleus
16	Isotopes have the same: (a) mass number (b) atomic number (c) number of neutrons (d) physical properties
17	The isotope used to treat cancer is: (a) cobalt-60 (b) carbon-14 (c) uranium-235 (d) iodine-131
18	Electronic configuration of sodium (Z = 11): (a) 2, 9 (b) 2, 8, 2 (c) 2, 8, 1 (d) 2, 1, 8
19	Formula of calcium hydroxide: (a) CaOH (b) Ca ₂ OH (c) CaOH ₂ (d) Ca(OH) ₂
20	Molecular mass of CaCO ₃ (Ca 40, C 12, O 16): (a) 100 (b) 68 (c) 84 (d) 116

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	b	Lesson 1
2	a	Lesson 1
3	d	$K = ^\circ C + 273$ (Lesson 1)
4	a	Lesson 2
5	d	Lesson 2
6	b	Lesson 2
7	b	Lesson 2 table
8	b	Lesson 3
9	c	Lesson 3
10	a	Lesson 3
11	a	Lesson 4
12	b	Lesson 5
13	b	$88 \div 44$ (Lesson 5)
14	a	Lesson 6
15	d	Lesson 6
16	b	Lesson 6
17	a	Lesson 6
18	c	Lesson 7
19	d	Lesson 8
20	a	$40 + 12 + 48$ (Lesson 8)

Quick checks

QC	Answers
1	a) Any two: camphor, naphthalene, iodine, ammonium chloride, dry ice b) Water seeps out through the pores and evaporates; evaporation cools c) 298 K
2	a) Mixture (an alloy of copper and zinc) b) Milk
3	a) Centrifugation b) Fractional distillation of liquid air
4	a) Chemical b) Physical c) Chemical
5	a) 44 g b) Lavoisier
6	a) 18 b) Iodine-131 c) James Chadwick
7	a) 2, 8, 2 b) 3
8	a) $Al_2(SO_4)_3$ b) 98

Day 8 recall (Block 0)

#	Answer	#	Answer
1	Newton	6	Convex
2	$\approx 9.8 \text{ m/s}^2$	7	2Ω
3	Concave	8	Velocity
4	$\approx 11.2 \text{ km/s}$	9	kWh (1 unit = $3.6 \times 10^6 \text{ J}$)
5	746 W	10	21% increase

Send me tonight

Day 8 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.